

Pentium Microprocessor

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Course Title: Microprocessors and Assembly Language
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Lecture References:

- ▶ Book:

- ▶ *Microprocessors and Interfacing: Programming and Hardware*, Chapter # 15, **Author:** Douglas V. Hall
- ▶ *The Intel Microprocessor 8086...Arch. Prog, Interfacing.* **Author:** Bary, Bray
- ▶ *Microprocessor and Microcomputer – Based System Design*, **Author:** Mohamed Rafiquzzaman

- ▶ Lecture Materials:

- ▶ *M.A. Sattar, BUET, Dhaka, Bangladesh.*

Introduction of Pentium

- ▶ The concepts of 80386 microprocessor and **80387 coprocessor** together evolved the 80486 microprocessor.
 - ▶ **Only new idea here is the introduction of 8K cache**
 - ▶ The cache is used for **storing both *data* and *instructions*** \
- ▶ Pentium was introduced **in 1993**
- ▶ Pentium processor is an improvement to the architecture found in 80486.

Introduction of Pentium

The Pentium, introduced in 1993, was similar to the 80386 and 80486 microprocess. It contained larger internal cache and data bus width is extended to 64-bit. The Table.1 shows the comparison between various pentium processors.

Processor	Released year	Data bus width	Memory size	L1 cache Data-Code	L2 cache	Bus transfer speed
Pentium 60 MHz 66 MHz 120 MHz 133 MHz 233 MHz	1993	64	4 GByte	8 K - 8 K	—	60 - 66 MHz
Pentium Pro 150-166 MHz	1995	64	64 GByte	8 K - 8 K	256 K	60 - 66 MHz

Pentium Improvements

- ▶ Improved cache structure
 - ▶ Reorganized to form **two level (L2) caches** that are each 8K bytes in size
 - ▶ One for **caching data**
 - ▶ Another for **caching instructions**
- ▶ Wider data bus width
 - ▶ Increased from **32 bit to 64 bits**
- ▶ Faster numeric processor
 - ▶ Operates about **5 times faster than the 80486** numeric processor

Pipelining in Pentium

- ▶ Pentium has two pipelines
- ▶ **U pipeline**
 - ▶ U-pipeline can execute **any Pentium instruction**
- ▶ **V pipeline**
 - ▶ V-pipeline only executes only **simple instructions**
- ▶ Each pipeline has 5 stages
 - ▶ i. Pre-fetch
 - ▶ ii. Instruction Decode
 - ▶ iii. Address Generation
 - ▶ iv. Execute, Cache, and ALU Access
 - ▶ v. Write back

Pipelining Stages in Pentium

▶ **Pre-fetch:**

- ▶ Instructions are fetched from the **Instruction cache** and **aligned in prefetch buffers for decoding**

▶ **Instruction Decode:**

- ▶ Instructions are decoded into the **Pentium's internal instruction format.**

▶ **Address Generation:**

- ▶ Address computations take place at this stage

▶ **Execute, Cache, and ALU Access:**

- ▶ The integer hardware executes the instruction

▶ **Write-back:**

- ▶ The results of the computation are written back to the register

Super Scalar Machine

- ▶ Any Processor capable of parallel instruction execution of multiple instructions is known as **superscalar machine**.
- ▶ As, in Pentium, there are two execution lines --- U-line and V-line, so Pentium is a ***Super Scalar Processor***
- ▶ What are the differences? to understand the concepts in terms of processor execution stages:
 - ▶ Parallelism
 - ▶ Simultaneity – Simultaneously Parallel
 - ▶ Pipelining – **Parallel and Simultaneous**

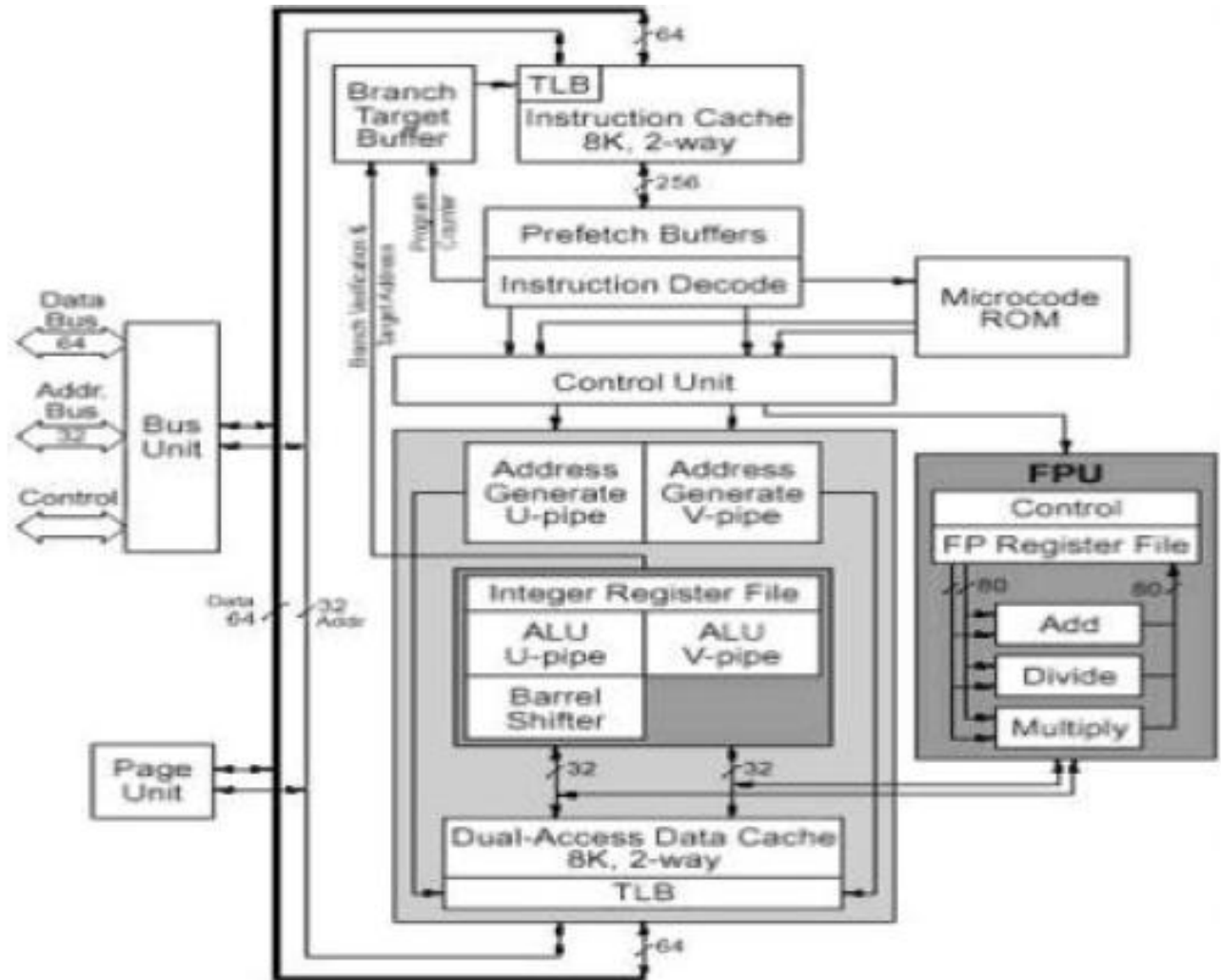
Pentium Architecture

It has all the units similar to 80386.

- Instruction unit
- Segmentation unit**
- Paging unit**
- Bus unit
- Execution unit

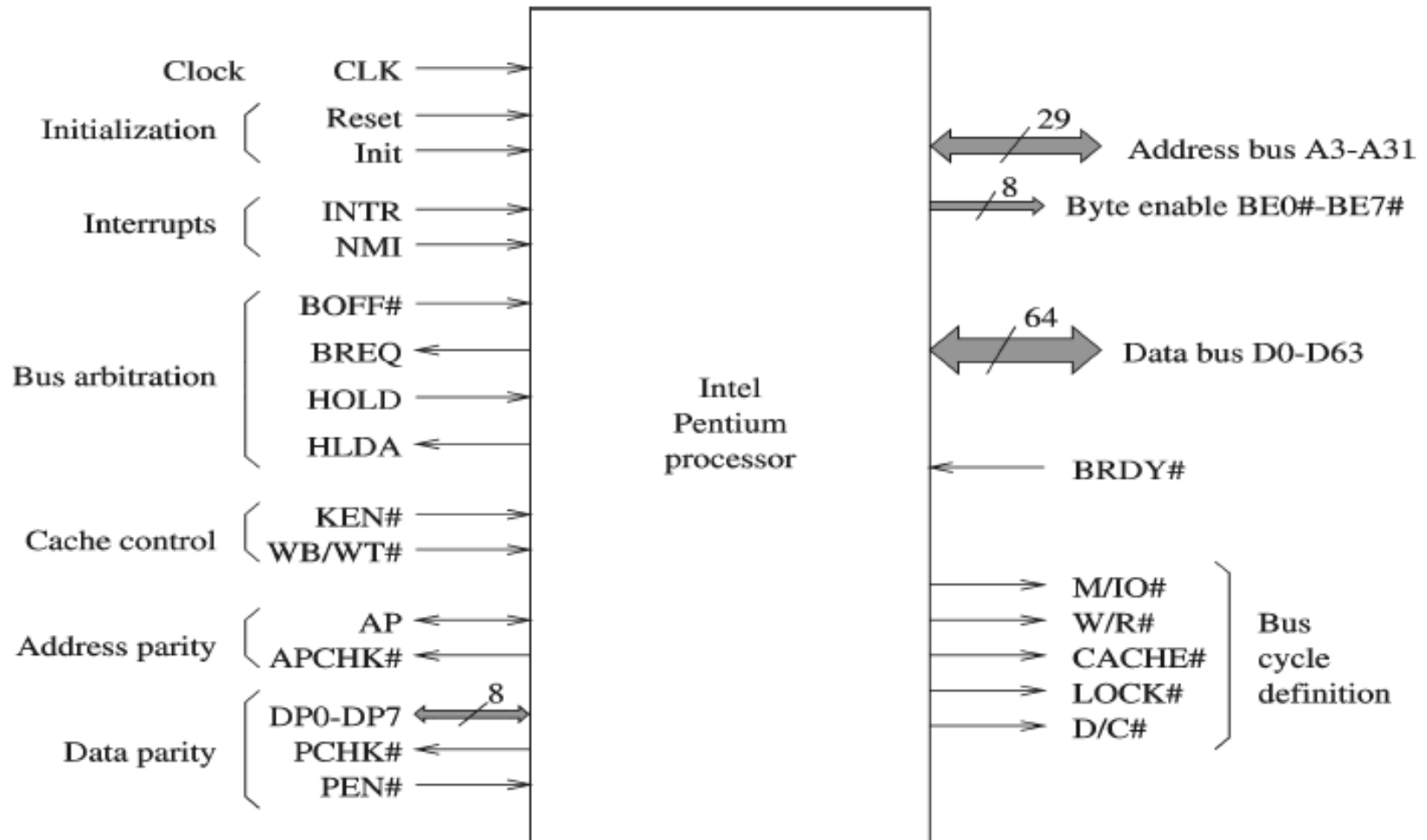
The newer one introduced as-

- Floating Point unit (FPU)**



Pentium Architecture

► Pin Diagram



Pentium Architecture

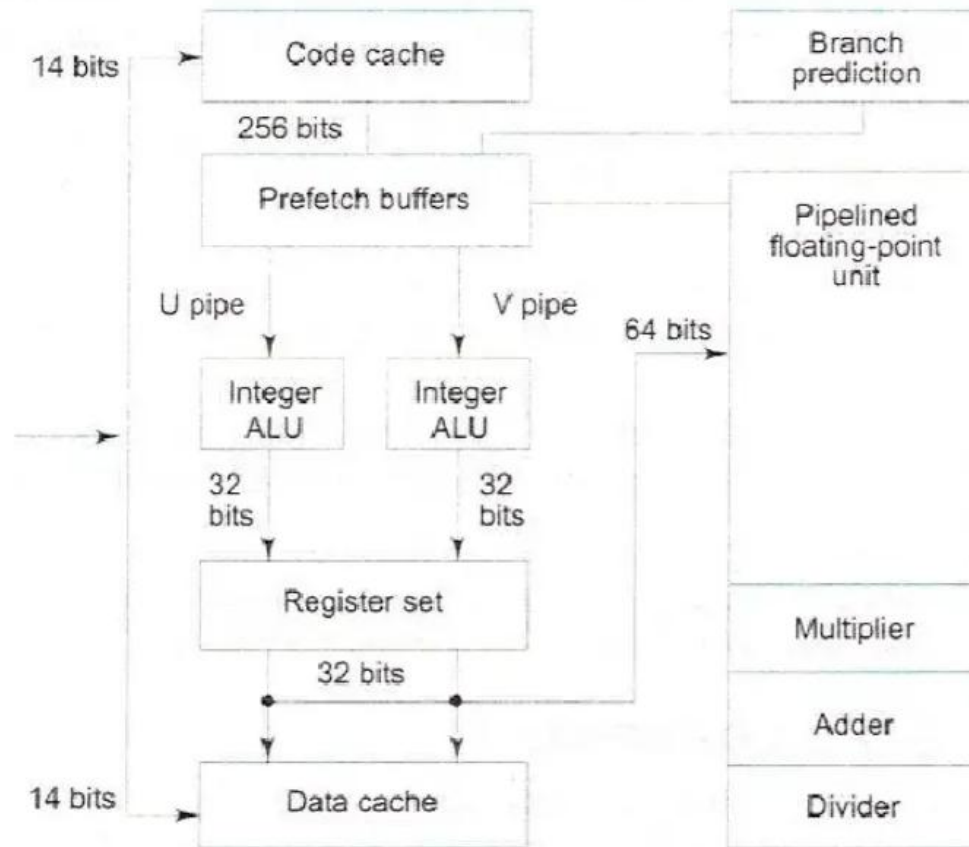


Fig. 11.1 Pentium CPU Architecture

Pentium Architecture:

Floating Point Unit (FPU)

- ▶ FPU has 8 80-bit general purpose floating point registers, ST(0) through ST(7)
- ▶ **It has 8-stage pipeline**
 - ▶ First 5 stages are identical to U and V pipelines
 - ▶ 2 additional execution stages
 - ▶ First execution stage (X1 stage) – For Sign Bit and Exponent Bits
 - ▶ Second execution stage (X2 stage) – For Mantissa/Fraction Values
 - ▶ In these two stages FPU reads the data from data cache and executes the floating point computation
- ▶ One additional error reporting stage

Pentium Architecture: **Pins**

- ▶ Packaged in 273 pins
 - ▶ Data Bus 64 pins
 - ▶ Address Bus 29 pins plus 3 pins
 - ▶ Control Bus 75 pins
 - ▶ Vcc + Ground 99 pins
 - ▶ No Connection (NC) 6 pins

Pentium Data Bus

- ▶ 64-bit data bus
- ▶ D0 through D63 (Data Bus)
- ▶ Bi-Directional
- ▶ These signals make up the Pentium's 64-bit bidirectional data bus.

Pentium Address Bus

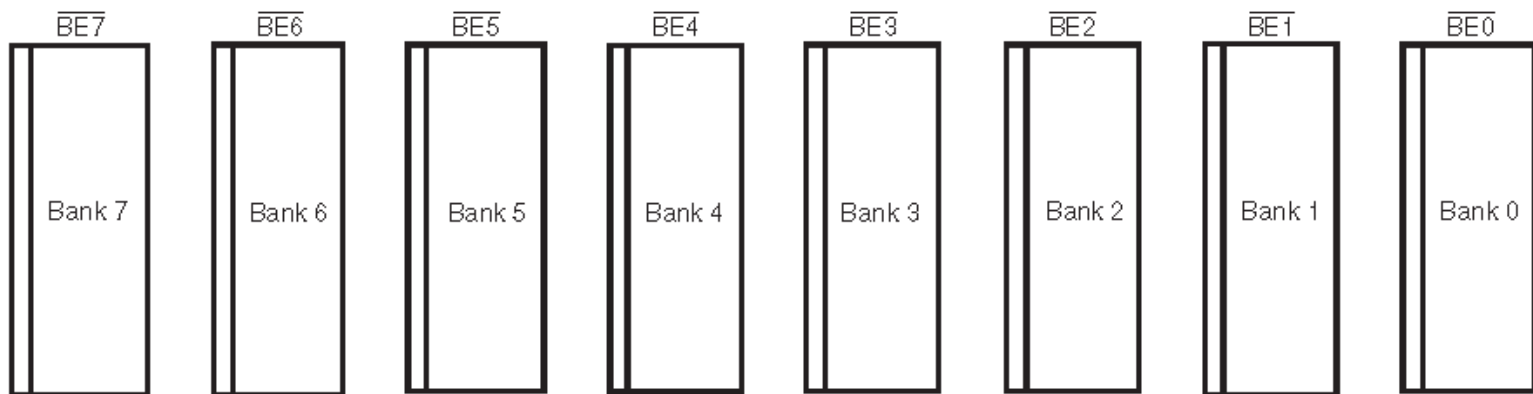
- ▶ 32-bit address bus.
- ▶ A3 through A31 (Address Lines)
 - ▶ Output/Input --- bi-directional
- ▶ These 29 address lines, together with the byte enable outputs *BE0-BE7*, form the Pentium's 32-bit address bus.
- ▶ A memory space of 4 GB is possible, along with 65536 I/O ports.
- ▶ The address lines are used as input during an inquire cycle to read an address *into the Pentium*, for examination by the internal cache.

Pentium Address Bus

- ▶ No A0, A1 and A2 address lines are available in Pentium
 - ▶ They are internally decoded to produce BE0 to BE7 signals
- ▶ **BE0 through BE7**
 - ▶ Byte (Bank???) Enable lines
 - ▶ Memory are arranged in total 8 banks
- ▶ **Output**
 - ▶ These, together with A3 through A31, make up the 32-bit address output by the Pentium.
 - ▶ Each byte enable is used to control a different 8-bit portion of the processor's 64-bit data bus.
 - ▶ BE0 enables Bank 0

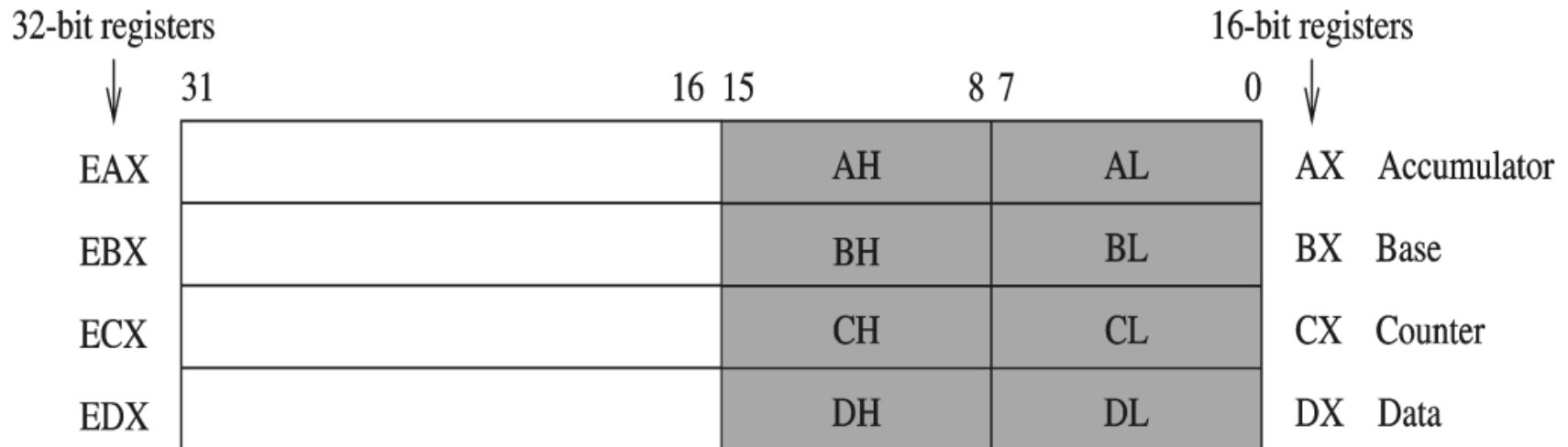
Pentium Memory System

- ▶ Pentium's memory arranged in 8 banks
- ▶ Each bank stores byte-wide data **with a parity bit**
 - ▶ It helps for **error detection and correction in data**
- ▶ BE0 to BE7 selects the banks
- ▶ New feature added to Pentium is its capability to **generate and check parity for address bus**



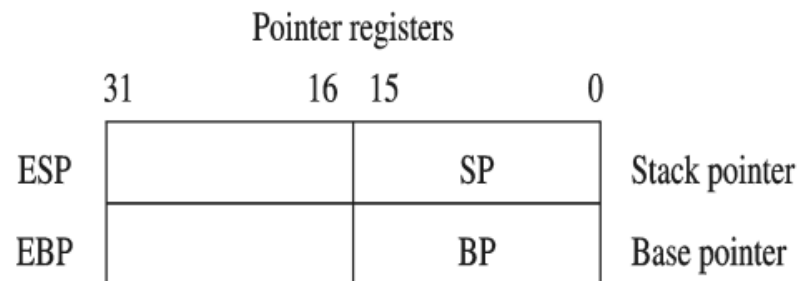
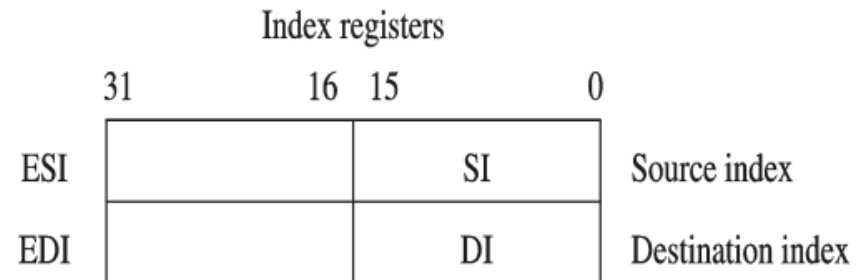
Pentium Registers

- Four 32-bit registers can be used as
 - * Four 32-bit register (EAX, EBX, ECX, EDX)
 - * Four 16-bit register (AX, BX, CX, DX)
 - * Eight 8-bit register (AH, AL, BH, BL, CH, CL, DH, DL)
- Some registers have special use
 - * ECX for count in loop instructions



Pentium Registers

- Two index registers
 - * 16- or 32-bit registers
 - * Used in string instructions
 - » Source (SI) and destination (DI)
 - * Can be used as general-purpose data registers
- Two pointer registers
 - * 16- or 32-bit registers
 - * Used exclusively to maintain the stack

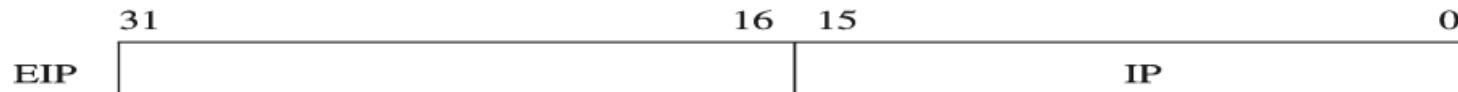


Pentium Registers

- Segment register

- * Six 16-bit registers
- * Support segmented memory architecture
- * At any time, only six segments are accessible
- * Segments contain distinct contents
 - » Code
 - » Data
 - » Stack

15		0
CS		Code segment
DS		Data segment
SS		Stack segment
ES		Extra segment
FS		Extra segment
GS		Extra segment



Features of All Pentium Processors



Pentium Pro: Salient Features

- ▶ The notable difference in the Pentium Pro than the earlier Pentium is that
 - ▶ There are provisions for a **36-bit address bus**, which allows access to **64G bytes of memory**.
 - ▶ ***This is meant for future use*** because no system today contains anywhere near that amount of memory.
- ▶ Pentium Pro is available in two versions.
 - ▶ One version contains a **256K level 2 cache**;
 - ▶ The other contains a **512K level 2 cache**
- ▶ Pentium Pro microprocessor is packaged in an immense **387-pin PGA** (pin grid array).

Pentium II: Salient Features

- ▶ Extension to Pro architecture with some differences
 - ▶ Internal cache in PII has been moved out of the chip
 - ▶ PII is not available as a single chip
 - ▶ Rather is available on a small plug-in circuit board, known as **Cartridge, along with level 2 (L2)** cache chip
- ▶ Various versions are available
 - ▶ **Celeron** is a version without L2 cache
 - ▶ **Xeon** is enhanced by having up to 2M L2 cache

Pentium III: Salient Features

- ▶ Based on Pro architecture, not on Pentium II
- ▶ Like PII, PIII is packaged in a cartridge instead of IC chip
- ▶ Additionally a **Coppermine** is packaged in an IC with 370 pins
- ▶ **Coppermine** is an internal cache with 256K advanced transfer mechanism within the IC running at processor speed
- ▶ **Why not used 512K Cache?**
 - ▶ *It has been observed that, increasing cache size from 256K to 512K improves the performance by only a few percent*

Pentium III: Salient Features

- ▶ Various versions of Pentium III are also available like Pentium II
 - ▶ **Standard** Pentium III
 - ▶ **Celeron** Pentium III uses 66MHz bus speed
 - ▶ **Xeon** Pentium III allows larger cache for server applications
 - ▶ Still Xeon is popular for server processors

Pentium IV: Salient Features

- ▶ Based on Pro architecture, not on P II or P III
- ▶ P IV is packaged in 421 pins IC
- ▶ It uses physically smaller transistors
 - ▶ Makes it much smaller and faster than P III
- ▶ Released initially in November 2000 with 1.3GHz speed
 - ▶ Was available with speed more than 3GHz

Thank You !!

